Master Thesis Research Roadmap: Adaptive LLM-Based Care Coach for Informal Caregivers in Switzerland

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Executive Summary

This research roadmap outlines a comprehensive master thesis project that extends the personality-aware chatbot framework from the preliminary study to develop an **Adaptive LLM-Based Care Coach** specifically designed for informal caregivers in Switzerland's home care system. The project builds upon the established Zurich Model-based personality detection and regulation system to create a specialized care coaching application that addresses the unique emotional, psychological, and practical needs of caregivers within the Swiss healthcare context.

Core Research Question

How can an adaptive LLM-based care coach, employing real-time personality detection and Zurich Model-guided behavior regulation, enhance resilience and reduce stress among informal caregivers in Switzerland's home care system?

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1. Research Foundation

1.1 Application Context: Swiss Informal Caregiving

Target Population: Approximately 700,000 Swiss residents (8% of population) serve as informal caregivers [ZHAW, 2024]

Context Characteristics:

Professional Training and Competency Gaps:
Unlike formal healthcare workers, informal caregivers typically lack systematic professional training, evidence-based caregiving techniques, or ongoing clinical supervision, leaving them to learn through trial-and-error in high-stakes situations. This professional training gap creates urgent needs for accessible, evidence-based practical guidance on caregiving techniques, emotional regulation strategies, crisis management protocols, and self-care practices adapted to their unique contexts and personality profiles.

Stress, Burnout, and Performance Impact:
Informal caregivers face chronic stress from prolonged caregiving responsibilities, leading to elevated burnout rates, emotional exhaustion, secondary traumatic stress, compassion fatigue, guilt, and isolation. These stress and negative mood states directly impair caregiving performance, decision-making quality, patient safety, and ability to provide consistent care, creating a vicious cycle where decreased performance increases guilt and stress, further deteriorating both caregiver wellbeing and care quality.

Vulnerable Population and Knowledge Deficits:
Informal caregivers constitute a vulnerable community exposed to exploitation, emotional manipulation, boundary violations, and systemic neglect, often sacrificing their own wellbeing without adequate recognition or support. Most caregivers lack critical knowledge on how to protect themselves from burnout, recognize warning signs of compassion fatigue, assert healthy boundaries, navigate legal protections, and access emergency support systems. Without professional training, caregivers struggle with establishing appropriate emotional and practical boundaries, leading to overextension, role confusion, caregiver-patient relationship strain, and family conflict.

System Complexity and Policy Navigation Challenges:
Swiss caregivers face strong cantonal variations in home care policies and Spitex service availability [MDPI, 2024], yet frequently remain unaware of their rights under Swiss law, available financial support mechanisms (cantonal allowances, caregiver leave policies), employment protection regulations, and healthcare system navigation strategies. This policy awareness gap compounds economic impacts, as many caregivers experience reduced working hours (avg. 8.4h/week reduction, ~CHF 970 monthly income impact) [PMC, 2023] without accessing support mechanisms that could mitigate financial strain.

Research Motivation:
While informal caregivers face documented stressors, existing digital support systems lack personality-adaptive emotional support and Swiss-specific healthcare knowledge navigation. This research develops a personality-aware chatbot providing: (1) emotional support tailored to individual personality profiles, and (2) accurate navigation of complex Swiss healthcare policies.

1.2 Existing Foundation: N8N Personality-Aware Chatbot workflow

The preliminary study developed a robust N8N-based workflow with:

Universal Components (applicable to all users): - **OCEAN personality detection**: Real-time assessment of Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism with Exponential Moving Average (EMA) smoothing for temporal stability - **Evidence scoring system**: Confidence scores quantifying the reliability of personality trait estimations based on conversation depth, linguistic evidence strength, and temporal consistency - **Zurich Model of Social Motivation mapping**: Identification of security, arousal, and affiliation motivational states derived from OCEAN profiles - **Persistent memory schema**: PostgreSQL-based storage system maintaining user session history,

personality profile evolution over time, conversation context, and longitudinal tracking data enabling temporal stability analysis and personalized adaptation across sessions

Caregiver-Specific Adaptations (new development): - **Context detection**: Economic stress, compassion fatigue, boundary challenges, Spitex navigation needs, policy/regulation knowledge gaps - **Regulation strategies**: Zurich Model-aligned behavioral directives tailored to caregiver emotional states and personality profiles - **Three-dimensional micro-coaching content**: - **(1) Emotional support**: Resilience building, stress validation, compassion fatigue recognition, boundary-setting - **(2) Educational support**: Practical caregiving techniques, time management, self-care strategies, crisis prevention training - **(3) Policy & regulation navigation**: Swiss healthcare system education (Spitex services, cantonal allowances, caregiver employment models, long-term care insurance, patient rights, advance directives), step-by-step application guidance, cost structure explanations

1.3 Research Gaps

Comprehensive literature analysis reveals interconnected deficiencies that this thesis systematically addresses:

Gap 1: Lack of Dynamic Personality-Aware Adaptation
Existing systems use static, rule-based personalization that ignores individual OCEAN personality traits and Zurich Model motivational states, resulting in suboptimal engagement and effectiveness. Furthermore, the field lacks standardized evaluation protocols or longitudinal tracking to assess technical performance and user acceptance of personality-adaptive interventions, hindering reproducibility and meta-analysis.

Gap 2: Zurich Model Theory-Practice Disconnect
No validated computational mappings translate Zurich Model of Social Motivation constructs into Zurich Model-aligned behavioral directives; measurement protocols to confirm intended motivational state induction remain undeveloped. The field lacks manipulation checks to confirm that adapted responses are perceived as intended by users.

Gap 3: LLM Safety, Reliability, and Privacy in Healthcare
Critical deficiencies exist in truthfulness, persona stability, and privacy preservation for vulnerable caregiver populations. No privacy-preserving architectures meet Swiss data protection standards while enabling effective personality tracking.

Gap 4: Absence of Personalized Education and Swiss Healthcare Navigation
No comprehensive caregiver support systems currently exist that integrate: (1) personalized practical education on caregiving techniques adapted to individual OCEAN personality profiles, and (2) LLM-based navigation of complex Swiss healthcare policies and regulations tailored to caregiver needs. While personality adaptation is essential for emotional support and educational guidance, the navigation of policies and regulations is fundamentally important functionality that does not necessarily require OCEAN trait

adaptation—instead, it requires accurate, contextualized information delivery. This gap is particularly acute in Switzerland's federalized system where caregivers require guidance on cantonal variations in Spitex services, caregiver allowances, long-term care insurance (Pflegeversicherung), family caregiver employment models, patient rights, and culturally appropriate Swiss interaction patterns. Current digital health interventions, when they exist, provide only generic information without personality adaptation or Swiss-specific contextualization.

Expected Impact:
By systematically addressing these four interconnected gaps, this research establishes a new paradigm for **theoretically grounded, personality-aware, safe, and contextually appropriate AI applications in informal caregiving**, with specific contributions to Switzerland's 700,000+ caregivers and broader implications for global caregiver support systems.

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2. Research Objectives and Questions

2.1 Primary Objectives

This thesis systematically addresses the four research gaps through interconnected objectives:

1. **Implement Dynamic Personality-Aware Adaptation with Rigorous Evaluation (Gap 1)**: Develop and validate OCEAN personality detection system with real-time adaptive coaching outperforming static approaches; establish standardized evaluation protocols using System Usability Scale (SUS) and user engagement metrics with longitudinal tracking (3-6 months) to assess system performance and user acceptance

2. **Operationalize Zurich Model with User-Validated Directives (Gap 2)**: Create validated computational mappings from personality profiles to Zurich Model behavior regulating directives with manipulation checks to confirm users perceive intended motivational states

3. **Ensure LLM Safety, Reliability, and Swiss Privacy Compliance (Gap 3)**: Design privacy-preserving architecture meeting Swiss data protection standards with persona stability mechanisms and hallucination guardrails

4. **Develop Three-Dimensional Personalized Education and Swiss Healthcare Navigation (Gap 4)**: Create comprehensive system integrating: (a) personality-adaptive emotional support and practical caregiving education tailored to individual OCEAN profiles, and (b) LLM-based Swiss healthcare policy and regulation navigation providing essential, accurate information on cantonal Spitex variations, caregiver allowances, long-term care insurance, patient rights, and cultural norms—recognizing that while personality adaptation enhances emotional and

educational support, policy navigation prioritizes accurate, contextualized information delivery

2.2 Research Questions

Primary Research Question

How can an adaptive LLM-based chatbot integrating real-time OCEAN personality detection with Zurich Model Zurich Model regulation deliver effective caregiver-specific micro-coaching for informal caregivers in Switzerland's healthcare system?

Sub-Research Questions

RQ1 — Personality Detection, Dynamic Adaptation, and Rigorous Evaluation (Gap 1)
How can real-time OCEAN detection combined with Zurich Model mapping enable dynamic micro-coaching adaptation that outperforms static approaches, and which standardized evaluation protocols rigorously assess system performance, user acceptance, and engagement?

RQ2 — Zurich Model Operationalization and Validation (Gap 2)
How can Zurich Model constructs be computationally operationalized into validated behavior regulating directives with user-confirmed perception of intended motivational states?

RQ3 — LLM Safety, Reliability, and Privacy (Gap 3)
What architectural and algorithmic strategies ensure LLM-based care coaching maintains truthfulness, persona stability, and privacy preservation for vulnerable caregivers?

RQ4 — Personalized Education and Swiss Healthcare Navigation (Gap 4)
How can practical caregiving education be personalized based on individual OCEAN personality profiles, while LLM-based Swiss healthcare policy and regulation navigation provides essential, accurate information delivery tailored to caregiver contexts, to address caregivers' comprehensive needs beyond emotional support?

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3. Methodology and System Architecture

3.1 N8N Pipeline Architecture

The system extends the existing N8N workflow with seven interconnected nodes:

Node 1: Enhanced Ingest & Context Analysis - Parse user input, validate session continuity - Extract caregiver context (caregiving situation, economic stress, emotional overload markers)

Node 2: Zurich Model Detection with EMA Smoothing (Gaps 1 & 2) - Real-time OCEAN trait estimation using LLM-based linguistic analysis - Retrieve personality history from PostgreSQL - Apply Exponential Moving Average (EMA) smoothing for temporal stability -

Map OCEAN traits to Zurich Model motivational domains (security, arousal, affiliation) - Track temporal stability (ICC >0.70 target) - Extract caregiver-specific features (compassion fatigue, guilt, boundary challenges)

Node 3: Behavioral Directive Generation (Gap 2) - Generate Zurich Model-aligned behavioral directives based on OCEAN profile and caregiver context - Weight directives according to motivational domains (security, arousal, affiliation) - Adapt to caregiver-specific emotional states (economic stress, burnout, isolation) - Generate regulation directives for response generation

Node 4: Evidence-Based Micro-Coaching Generation (Gap 4) - Construct LLM prompt integrating: - User message and history - OCEAN profile and Zurich Model-aligned behavioral directives - Caregiver-specific instructions (economic stress, compassion fatigue, boundary-setting, Spitex navigation) - Swiss healthcare context (cantonal policies, cultural norms) - Generate response using GPT-4 or open-source alternative (Llama/Mistral) - Implement hallucination guardrails: - Retrieval-augmented generation for Swiss policy facts - Uncertainty disclosure when confidence is low

Node 5: Persona Stability & Response Verification (Gap 3) - Validate persona stability: Empathetic care coach identity maintained without false credentials or inappropriate boundaries - Verify response grounding and appropriateness

Node 6: PostgreSQL Persistence & Longitudinal Tracking (Gap 6) - Save session metadata: user ID, timestamp, OCEAN estimates, Zurich Model weights - Save conversation turn: user/system messages, context, safety flags - Update personality history for temporal tracking - Log implementation outcomes: latency, costs, errors

Node 7: Response Delivery - Return micro-coaching response via API - Log response for system monitoring and evaluation

3.2 Privacy and Safety Architecture (Gap 3)

Privacy Protections: - Data encryption: PostgreSQL at rest (AES-256), API over HTTPS/TLS - Access controls: Role-based permissions, researcher-only identifiable data access - Anonymization: Pseudonymized user IDs, separated personally identifiable information - Audit logging: All data access events logged - Swiss GDPR compliance: Informed consent, data retention (12 months), right-to-deletion

Safety Protocols: - Response appropriateness validation - Human oversight: Expert review of concerning sessions

3.3 Caregiver-Specific Micro-Coaching Library (Gap 4)

The micro-coaching system provides **three integrated support dimensions**: (1) emotional resilience building with personality adaptation, (2) practical education on caregiving strategies personalized to OCEAN profiles, and (3) RAG-based navigation of Swiss

healthcare policies and regulations. This multi-dimensional approach addresses the comprehensive needs of informal caregivers beyond traditional emotional support.

Dimension 1: Emotional Support and Resilience Building

The system provides personality-adaptive emotional support addressing compassion fatigue recognition, stress validation and coping strategies, and boundary-setting guidance. Content is dynamically adapted based on individual OCEAN personality profiles and Zurich Model motivational states to ensure resonance with each caregiver's psychological needs.

Dimension 2: Educational Support and Caregiving Skills

Practical caregiving education is personalized to individual OCEAN traits, covering evidence-based caregiving techniques, time management and organization strategies, self-care education, and crisis prevention training. Educational content adapts to personality profiles to optimize learning engagement and practical application.

Dimension 3: Policy, Regulation, and Healthcare Navigation

A Retrieval-Augmented Generation (RAG) system provides accurate, grounded information on Swiss healthcare policies and regulations, including: economic stress and financial support programs (cantonal caregiver allowances, family caregiver employment models), Swiss Spitex system navigation (service explanation, canton-specific coverage, respite care options, cost structures), cantonal healthcare policy education (federalized system variations, long-term care insurance, patient advocacy), regulatory compliance and legal guidance (patient rights, advance directives, caregiver leave policies), and healthcare system navigation (care coordination, insurance coverage, community resources). The RAG system ensures factual accuracy by retrieving information from curated Swiss healthcare policy databases, preventing hallucinations and providing contextualized guidance based on the caregiver's specific canton and caregiving situation.

3.4 Comprehensive Evaluation Framework

3.4.1 Primary Outcome Measures

System Usability: System usability is assessed using the System Usability Scale (SUS) at 2 weeks and completion, targeting scores above 68 (above average) and ideally exceeding 80.3 (excellent). The Technology Acceptance Model (TAM) measures perceived usefulness and ease of use.

User Engagement: User engagement metrics include session frequency (targeting 3-7 sessions per week), session duration (targeting 5-15 minutes), completion rates, and longitudinal retention tracked at 4 weeks, 8 weeks, 3 months, and 6 months. User satisfaction ratings and qualitative feedback complement these quantitative measures.

Perceived Helpfulness: Perceived helpfulness is evaluated through post-session helpfulness ratings on a 7-point scale, qualitative feedback assessing personality

adaptation appropriateness, and self-reported behavioral changes including coping strategy adoption.

3.4.2 Manipulation Checks

Personality Detection Validation (Gap 1): Personality detection accuracy is validated through expert annotation agreement requiring at least 85% concordance between system predictions and 3-5 expert psychologists across 30 sessions. Temporal stability is assessed using Intraclass Correlation Coefficient (ICC) exceeding 0.70 across sessions, while self-report correlation with the Big Five Inventory (BFI-44) must achieve $r>0.40$.

Zurich Model Induction Confirmation (Gap 2): Zurich Model state induction is confirmed through Ecological Momentary Assessment using post-session ratings of perceived security, arousal, and affiliation on 7-point scales. Content analysis verifies that targeted Zurich Model elements are present in generated responses, while user perception analysis confirms that adapted responses align with intended motivational states.

3.4.3 Implementation and Fairness Outcomes

Cost-Effectiveness: Cost-effectiveness analysis tracks development cost per user and operational cost per session to assess economic viability.

Adoption: Adoption metrics include recruitment rate (targeting above 10%) and onboarding completion (targeting above 80%). Technical reliability is monitored through system uptime exceeding 95%, error rate below 5%, and response latency under 3 seconds.

Safety: Safety monitoring tracks incident rates per 1,000 interactions and any adverse events reported by users.

Fairness Auditing: Fairness is assessed through stratified outcome analysis across demographic groups including age (<50, 50-65, >65), gender, education level, digital literacy, linguistic region, and canton. Interaction effect testing examines whether system effectiveness varies across these demographic groups, while engagement equity analysis ensures comparable session frequency, duration, and completion rates across all user segments.

3.4.4 Longitudinal Tracking

Follow-Up Schedule: The longitudinal tracking protocol includes an intensive phase spanning weeks 0-8 with full system access and biweekly user satisfaction and engagement assessments. Post-intervention follow-ups occur at 3 months and 6 months to assess durability after system withdrawal. Relapse indicators track the proportion of users returning to baseline levels and identify predictors of sustained improvement.

Qualitative Mixed-Methods: Qualitative insights are gathered through semi-structured interviews with 10-15 participants exploring user experience, caregiver-specific value, cultural appropriateness, and improvement suggestions. Expert evaluation involves 3-5

Swiss healthcare professionals assessing therapeutic alignment, safety considerations, and cantonal relevance. Thematic analysis employs grounded theory coding to identify emergent themes regarding system acceptability, implementation barriers, and facilitators of adoption.

3.4.5 Reproducibility Standards

Reproducibility is ensured through protocol preregistration on the Open Science Framework (OSF) with detailed analysis plans specifying statistical tests, effect size calculations, and sample size justifications. Anonymized data will be shared via restricted-access repositories requiring ethical approval. The study adheres to CONSORT-EHEALTH reporting standards for digital health interventions to maximize transparency and reproducibility.

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4. Implementation Phases

Phase 1: Foundation and Expert Consultation (Weeks 1-3)

Objective: Establish theoretical foundations and validate design with Swiss healthcare experts

Deliverables: - Comprehensive literature synthesis integrating seven survey papers with Swiss caregiver research - Zurich Model computational mappings: OCEAN→Zurich Model-aligned behavioral directive weight tables validated by experts - Swiss contextualization framework: Cantonal policy database, cultural interaction patterns - Enhanced personality detection protocol: Caregiver-specific feature extraction, expert annotation guidelines

Key Activities: - Review 20-30 key papers on caregiver psychology, digital health, and Swiss healthcare policy, supplemented by 5-7 systematic review papers synthesizing broader literature domains - Conduct expert interviews (n=3-5): Swiss geriatric psychologists, home care policy experts, Spitex coordinators - Validate Zurich Model-aligned behavioral directive mappings for caregiver appropriateness - Compile cantonal policy database (26 Swiss cantons: funding models, Spitex coverage, employment regulations) - Establish annotation protocol: 3-5 experts annotate 30 simulated conversations; target ≥85% inter-annotator agreement

Phase 2: System Development (Weeks 4-9)

Objective: Implement N8N workflow with privacy-preserving architecture and caregiver-specific components

Deliverables: - Privacy-preserving N8N workflow: 7-node pipeline with encrypted PostgreSQL, access controls, audit logging - Dynamic personality detection module: OCEAN + caregiver emotional state assessment with EMA smoothing - Zurich Model

regulation engine: Directive weighting system - Evidence-based micro-coaching library: Economic stress, compassion fatigue, boundary-setting, Spitex navigation templates

Key Activities: - Develop N8N workflow nodes (context analysis, personality detection, Zurich Model regulation, micro-coaching generation, safety checks, PostgreSQL persistence) - Implement privacy architecture: Data encryption, access controls, anonymization, GDPR compliance - Create caregiver intervention library validated by Swiss professionals - Configure PostgreSQL schemas: Sessions, personality history, conversation turns - Integrate safety protocols: Response appropriateness validation, persona stability checks

Phase 3: Validation and Pilot Deployment (Weeks 10-13)

Objective: Systematically validate research gap solutions through simulations, expert reviews, and pilot caregiver deployment

Deliverables: - Personality detection validation: Expert agreement, temporal stability ICC, BFI correlation - Zurich Model manipulation check results: Perception validation confirming targeted state induction - Safety protocol validation: Response appropriateness assessment, persona stability evaluation, privacy compliance audit - Caregiver implementation evidence: Dose-response patterns, component effectiveness, equity assessment - Swiss contextualization evidence: Cantonal adaptation functionality, cultural appropriateness evaluation - Standardized outcome assessment: Pre-post SUS scores, engagement metrics, user satisfaction, qualitative findings, cost analysis

Key Activities:

Controlled Simulation Studies: - Create 30 simulated caregiver personas: OCEAN diversity, caregiving contexts, Swiss regional variations - Expert validation: 3-5 psychologists annotate OCEAN profiles; evaluate Zurich Model-aligned behavioral directive appropriateness - Safety testing: Verify response appropriateness and persona stability across diverse scenarios

Expert Evaluation: - Academic supervisor review: Weekly meetings, methodology validation - Swiss mental health professionals (n=3-5): Therapeutic alignment, cultural appropriateness, safety review - Home care policy experts (n=2-3): Cantonal policy accuracy, Spitex information verification

Pilot Caregiver Deployment (n=10-15 participants): - **Inclusion criteria**: ≥18 years, ≥10 hours/week informal care, device access - **Baseline assessment**: Demographics, BFI-44 (personality baseline) - **Intensive phase (4 weeks)**: Target 3-7 sessions/week; periodic Zurich Model perception validation surveys - **Post-intervention assessment**: SUS, TAM, user satisfaction ratings, engagement metrics, qualitative feedback on system appropriateness - **Semi-structured interviews (all participants)**: User experience, caregiver-specific value, cultural appropriateness, safety perceptions, improvement suggestions - **Fairness auditing**: Stratified outcome analysis by age, gender, education, digital literacy, canton - **Performance analysis**: Calculate expert agreement, ICC, BFI

correlation, Zurich Model perception validation, effect sizes (Cohen's d), SUS scores, cost per user, technical reliability

Phase 4: Robustness Testing and Longitudinal Follow-Up (Weeks 14-17)

Objective: Enhance robustness through adversarial testing, refine based on findings, initiate 3-month follow-up for durability assessment

Deliverables: - Comprehensive failure-mode catalog: Taxonomy of hallucinations, prompt injections, safety errors with mitigation strategies - Refined guardrail suite: Enhanced safety prompts, improved content filters - Robustness benchmarks: Performance under adversarial conditions (jailbreaks, long contexts, multilingual edge cases, noisy input) - 3-month follow-up initiation: Assess sustained engagement patterns and long-term user satisfaction - Deployment readiness documentation: Monitoring dashboards, incident playbooks, GDPR compliance verification, implementation guide

Key Activities:

Red-Team Adversarial Testing: - **Failure mode discovery**: Prompt injection attacks, jailbreak scenarios, hallucination provocation, boundary violation attempts - **Mitigation implementation**: Content filtering, grounding enforcement (RAG for Swiss policies), uncertainty disclosure, persona stability guardrails - **Robustness stress testing**: Long conversations (50+ turns), noisy input (typos, emotional expressions), concurrent sessions (load testing), error recovery (API failures)

System Refinement: - Adjust personality detection if expert agreement <85%, temporal stability ICC <0.70, or BFI correlation <0.40 - Optimize Zurich Model regulation if perception validation scores <0.50; review qualitative feedback on directive helpfulness - Prioritize effective caregiver components based on effect sizes and user satisfaction - Update cantonal policy database based on expert/participant feedback

3-Month Longitudinal Follow-Up (Week ~22-23): - Contact pilot participants for follow-up user satisfaction and system usage assessments - Calculate proportion maintaining ≥50% of initial improvement - Analyze predictors: personality profiles, engagement patterns, caregiver contexts - Assess sustained behavior change: Continued use of learned coping strategies, boundary-setting, self-care

Phase 5: Thesis Writing and Dissemination (Weeks 18-25)

Objective: Complete comprehensive thesis, prepare academic dissemination, enable technology transfer

Deliverables: - Master thesis manuscript: Research gap solutions with theoretical contributions, empirical evidence, implementation guidance - Open science release: GitHub repository with N8N workflows, PostgreSQL schemas, evaluation protocols, anonymized datasets, analysis code - Implementation toolkit: Deployment guide, cost

calculator, training materials, ethical frameworks for Swiss Spitex/healthcare organizations

Thesis Structure:

Chapter 1: Introduction - Application context: Swiss informal caregivers (700,000+ population) facing emotional and navigational challenges - Research gaps from literature synthesis - Primary research question and sub-research questions

Chapter 2: Literature Review and Research Gaps - Synthesize seven survey papers: Caregiver micro-coaching, Zurich Model chatbots, LLM care coaches, personality-adaptive systems - Document research gaps: Personality adaptation, Zurich Model operationalization, LLM safety, personalized education and Swiss healthcare navigation - Position thesis as systematic gap-filling

Chapter 3: Methodology and System Design - Dynamic personality detection architecture (Gap 1) - Zurich Model operationalization (Gap 2) - Privacy-preserving architecture and safety protocols (Gap 3) - Caregiver-specific micro-coaching library (Gap 4) - Swiss contextualization (Gap 5) - Comprehensive evaluation framework (Gap 6)

Chapter 4: Implementation and Validation - System development narrative (Phases 1-2) - Simulation studies: Personality detection validation, Zurich Model-aligned behavioral directive appropriateness, safety testing - Pilot deployment: Demographics, intervention fidelity, engagement patterns - Adversarial testing: Failure modes, robustness benchmarks, mitigations

Chapter 5: Results - RQ1: Personality detection accuracy (agreement, ICC, BFI correlation); moderation analysis - RQ2: Zurich Model manipulation checks (perception validation); mediation analysis - RQ3: Safety performance (response appropriateness); privacy compliance - RQ4: Caregiver outcomes (user engagement, satisfaction, system usability, dose-response, component effectiveness, equity) - RQ5: Swiss contextualization evidence (cantonal functionality, cultural appropriateness) - RQ6: Standardized outcomes (SUS, engagement, implementation metrics); qualitative themes - Longitudinal: 3-month durability (proportion maintaining gains, relapse predictors); usage trajectories

Chapter 6: Discussion - Interpretation: How results address each gap; comparison to prior literature - Theoretical contributions: Personality psychology (OCEAN-Zurich Model interactions), HCI (persona stability-adaptation tradeoffs), implementation science (resource-constrained deployment) - Practical implications: Caregiver support scalability, Swiss healthcare integration, policy recommendations - Limitations: Small sample (n=10-15), 3-6 month follow-up insufficient for long-term assessment - Ethical considerations: Automated support boundaries, informed consent, privacy, dependency risks, human oversight - Future directions: RCT with control group, multilingual expansion, cantonal-specific adaptations, EHR/Spitex integration

Chapter 7: Conclusion - Reiterate research gaps and systematic solutions - Summarize key contributions: Personality-aware framework, Zurich Model operationalization, safety

protocols, three-dimensional personalized education framework, Swiss contextualization - Emphasize impact: Scalable emotional support and knowledge navigation system for Swiss caregivers, replicable methodology for vulnerable populations, open-science enablement

Open Science Release: - GitHub repository: N8N workflow JSONs, PostgreSQL schemas, evaluation protocols, anonymized datasets, R/Python analysis scripts, documentation - OSF: Preregistered protocol, analysis plan, results (including null findings), deviations

Implementation Toolkit: - Deployment decision framework: Organizational readiness assessment, target population identification, integration planning - Cost-effectiveness calculator: User count, API costs, server hosting → cost per user analysis - Training materials: Caregiver onboarding video/manual/FAQ, staff monitoring dashboard tutorial, incident response training, ethical boundaries workshop - Ethical documentation: Informed consent template, GDPR checklist, privacy protections, appropriate boundaries, human oversight requirements

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5. Expected Contributions and Impact

5.1 Academic Contributions Aligned with Research Gaps

Contribution 1: Dynamic Personality-Aware Adaptation Framework (Gap 1)
First operational system integrating real-time OCEAN detection with Zurich Model motivational states for caregiver micro-coaching. Demonstrates dynamic adaptation outperforms static personalization through validated detection (>85% expert agreement) and EMA smoothing. Establishes evidence for how personality differences moderate intervention effectiveness, providing foundational framework for future personality-adaptive AI for caregiver support.

Contribution 2: Validated Zurich Model Operationalization (Gap 2)
Creates first validated computational mappings translating Zurich Model constructs into behavior regulating directives with confirmed measurement protocols. Implements validation methods confirming targeted motivational states are induced. Provides preliminary causal evidence linking motivational state modulation to resilience outcomes via mediation analysis. Advances the integration between Zurich Model psychological theory and validated computational implementation for adaptive conversational AI.

Contribution 3: Safety and Reliability Framework for Caregiver Support LLMs (Gap 3)
Develops comprehensive privacy-preserving architecture with persona stability mechanisms and hallucination guardrails for vulnerable caregivers. Provides reproducible safety framework for responsible AI deployment in caregiver support applications.

Contribution 4: Three-Dimensional Personalized Education and Guidance Framework (Gap 4)
Pioneers **three-dimensional micro-coaching framework** that extends beyond traditional

emotional support to integrate: (1) emotional support (resilience, compassion fatigue recognition, boundary-setting) with personality adaptation based on OCEAN traits, (2) practical caregiving education (caregiving techniques, time management, self-care strategies, crisis prevention) personalized to individual OCEAN personality profiles, and (3) LLM-based Swiss healthcare policy and regulation navigation (Spitex system, cantonal allowances, caregiver employment models, long-term care insurance, patient rights) providing essential, accurate information delivery tailored to caregiver contexts (e.g., Alzheimer's vs. disability care). Demonstrates the strategic integration of personality-adaptive content (for emotional and educational support) with essential information delivery (for policy navigation), establishing first framework for multi-dimensional personalized caregiver support that recognizes when personality adaptation is beneficial versus when accurate, contextualized information is the priority.

Contribution 5: Swiss Healthcare Contextualization Framework (Gap 5)
First personality-adaptive care coaching system designed for Switzerland's unique federalized healthcare system, demonstrating systematic adaptation to cantonal variations, Spitex integration, and culturally sensitive interaction patterns. Establishes replicable methodology for contextualizing international AI research to regional healthcare systems with federalized structures.

Contribution 6: Standardized Evaluation Protocols and Longitudinal Evidence (Gap 6)
Implements first standardized evaluation framework using system usability measures (SUS), user engagement metrics, manipulation checks, and fairness auditing with longitudinal tracking (3-6 months). Documents implementation outcomes (cost per user, adoption rates) enabling cost-effectiveness assessments. Establishes reproducible evaluation standards and core outcome sets enabling meta-analyses.

5.2 Theoretical Contributions

Personality Psychology: Advances understanding of OCEAN-Zurich Model interactions in healthcare, providing evidence for optimal directive weighting across personality profiles.

Human-Computer Interaction: Extends HCI theory on personalized conversational agents by demonstrating measurable advantages of dynamic adaptation over static approaches in vulnerable populations.

Implementation Science: Contributes methodological framework for translating personality-adaptive AI from research prototypes to practice-ready systems within constrained resources using open-source technologies.

5.3 Practical Impact

Direct Caregiver Support: Scalable, evidence-based personality-adaptive system providing: (1) emotional support for compassion fatigue, boundary challenges, and isolation, and (2) knowledge navigation for complex Swiss healthcare policies. Enables 24/7 accessible psychological support and information guidance complementing limited

formal mental health services. Note: The system addresses emotional and informational needs but does not resolve economic challenges caregivers may face.

Healthcare System Benefits: Reduces caregiver burnout and improves resilience, potentially decreasing premature institutionalization and healthcare costs. Supports Switzerland's innovative family caregiver employment models.

Technology Transfer: Open-source release enables Swiss healthcare organizations, Spitex services, and international research community to build upon this work.

Policy Implications: Establishes evidence-based foundation for Swiss policy discussions on AI integration in caregiver support programs. Demonstrates feasibility of culturally sensitive, privacy-preserving, safe personality-adaptive AI meeting Swiss regulatory requirements.

Global Generalizability: While contextualized for Switzerland, the systematic methodology provides replicable framework for other countries developing personality-adaptive AI for caregiver support with federalized structures, linguistic diversity, and unique cultural norms.

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6. Risk Assessment and Mitigation

6.1 Technical Risks

Risk	Likelihood	Impact	Mitigation Strategy
Personality detection accuracy below threshold	Medium	High	Extensive expert validation; adjust detection parameters; augment training data
LLM hallucinations in health guidance	Medium	High	RAG for Swiss policies; uncertainty disclosure; content filtering
Deployment infrastructure and performance latency	Medium	High	Optimize N8N workflow efficiency; use efficient open-source LLMs (Ollama with quantized models); implement caching for personality profiles; set response timeout thresholds
Insufficient funding for high-performance cloud computing	Medium	Medium	Prioritize local deployment on university servers; leverage free-tier cloud services; use open-source models requiring lower computational resources; implement efficient prompt engineering to reduce API costs
System scalability under load	Low	Medium	Cloud architecture; load testing; graceful degradation

Risk	Likelihood	Impact	Mitigation Strategy
Privacy/data security breach	Low	High	Encryption, access controls, audit logging, GDPR compliance, penetration testing

6.2 Research Risks

Risk	Likelihood	Impact	Mitigation Strategy
Limited caregiver participation (n<10)	Medium	High	Multiple recruitment channels (Spitex, support organizations, online forums); early outreach; flexible participation (remote, asynchronous)
Ethical concerns about AI emotional support	Low	High	Clear boundaries, professional oversight, transparent disclosure of AI nature and limitations
Cantonal variation complexity exceeds scope	Medium	Medium	Focus on primary cantons (Zurich, Bern, Geneva); document variations without full implementation

6.3 Timeline Risks

Risk	Likelihood	Impact	Mitigation Strategy
Development delays (Phases 1-2)	Medium	Medium	Agile development with regular milestones; prioritize core features; leverage existing N8N foundation
Recruitment delays (Phase 3)	Medium	High	Early recruitment planning (Week 8); multiple strategies; backup simulated scenarios
LLM API cost overruns	Low	Medium	Budget monitoring; implement open-source alternatives (Ollama); optimize prompt efficiency
Expert availability constraints	Low	Medium	Early engagement; flexible interview scheduling; remote consultations

7. Resource Requirements

7.1 Technical Resources (Open-Source Priority)

- **Computing**: University computing resources or free-tier cloud services
- **Software**: N8N Community Edition (self-hosted), PostgreSQL, Python, Node.js
- **APIs**: University-provided LLM access or open-source alternatives (Llama, Mistral via Ollama)

- **Storage**: Local or university secure storage for interaction logs
- **Language**: Open-source translation tools for Swiss language support as needed

7.2 Human Resources (Volunteer-Based)

- **Supervision**: Regular meetings with academic supervisors (standard thesis support)
- **Expert consultation**: Voluntary review from academic network (Swiss psychologists, home care experts, Spitex coordinators)
- **Caregiver participants**: 10-15 participants recruited via Swiss caregiver organizations, Spitex networks
- **Development**: Student researcher (thesis author)

7.3 Time Allocation (25-week thesis)

- **Phase 1** (Weeks 1-3): Foundation and expert consultation
- **Phase 2** (Weeks 4-9): System development
- **Phase 3** (Weeks 10-13): Validation and pilot deployment
- **Phase 4** (Weeks 14-17): Robustness testing and 3-month follow-up initiation
- **Phase 5** (Weeks 18-25+): Thesis writing, 6-month follow-up, dissemination

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8. Ethical Considerations

8.1 Informed Consent and Transparency

- Clear disclosure: AI nature, research purpose, data use, voluntary participation, withdrawal rights
- Explicit limitations: Not substitute for professional mental health care; cannot diagnose or prescribe
- Informed consent template meeting Swiss ethical standards

8.2 Privacy and Data Protection

- Swiss GDPR compliance: Data minimization, purpose limitation, storage limitation (12 months)
- Encryption: Data at rest (AES-256), data in transit (HTTPS/TLS)
- Anonymization: Pseudonymized IDs, separated personally identifiable information
- Right-to-access, right-to-deletion, data portability mechanisms

8.3 Safety and Appropriate Boundaries

- Appropriate boundaries: Care coach identity without false therapeutic credentials; emotional support not therapy
- Response validation: Ensure responses maintain empathetic care coach persona while avoiding inappropriate advice

- Human oversight: Expert review of flagged sessions, incident documentation

8.4 Equity and Fairness

- Fairness auditing: Stratified outcome analysis ensuring equitable effectiveness across age, gender, education, digital literacy, SES, canton
- Accessibility: Design for low digital literacy; provide onboarding support
- Cultural sensitivity: Swiss caregiving norms, privacy expectations, multilingual considerations

8.5 Dependency and Over-Reliance Risks

- Monitor usage patterns: Assess whether users maintain voluntary engagement or show dependency signs
- Empower self-management: Teach coping strategies enabling autonomy rather than system reliance
- Planned discontinuation: 3-6 month follow-ups assess whether users maintain gains without continued system access

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9. Timeline and Milestones

Week	Phase	Key Milestones	Deliverables
1-3	Phase 1	Literature synthesis complete; Expert interviews conducted; Zurich mappings validated	Computational mappings, Swiss policy database, annotation protocol
4-9	Phase 2	N8N workflow operational; PostgreSQL schemas deployed; Safety protocols integrated	Functional care coach system, intervention library, privacy architecture
10-13	Phase 3	Simulation validation complete; Pilot deployment launched (n=10-15); Baseline/post-intervention data collected	Validation report, pilot outcomes, qualitative interview data
14-17	Phase 4	Adversarial testing complete; System refinements deployed; 3-month follow-up initiated	Failure-mode catalog, robustness benchmarks, refined system
18-25	Phase 5	Thesis chapters drafted; Results analyzed; 6-month follow-up completed	Master thesis manuscript, implementation toolkit, GitHub release

Key Decision Points: - **Week 3**: Expert validation of Zurich mappings → Proceed to development or refine mappings - **Week 9**: System functional testing → Proceed to pilot or extend development - **Week 13**: Pilot preliminary results → Assess effectiveness evidence;

decide on refinements - **Week 17**: Adversarial testing outcomes → Finalize system version for documentation - **Week 25**: Thesis defense preparation; Open science release

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10. Success Criteria

10.1 Technical Success Criteria

- Personality detection: ≥85% expert agreement, ICC >0.70, r>0.40 with BFI-44
- Zurich Model manipulation: Perception validation scores >0.50 between intended directives and perceived states
- Safety performance: Response appropriateness validation, persona stability maintenance across sessions
- System reliability: Uptime >95%, error rate <5%, response latency <3 seconds

10.2 Clinical Effectiveness Criteria

- User satisfaction: Mean helpfulness rating ≥4.0/5.0
- Engagement quality: Meaningful interaction depth and sustained usage patterns
- Engagement: ≥70% participants complete 4-week intervention; mean ≥3 sessions/week
- Usability: Mean SUS score >68 (above average)
- Longitudinal durability: ≥50% of participants maintain ≥50% of gains at 3-month follow-up

10.3 Academic Contribution Criteria

- Research gaps systematically addressed with empirical evidence
- Reproducible methodology documented enabling replication
- Open science release: Code, data, protocols publicly available

10.4 Practical Impact Criteria

- Implementation toolkit created for Swiss healthcare organizations
- Cost-effectiveness analysis completed: Cost per user, operational efficiency metrics
- Swiss expert endorsement: Positive evaluation from ≥3 Swiss mental health/home care professionals
- Scalability assessment: Documented feasibility for deployment to larger caregiver populations

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11. Conclusion

This research roadmap presents a comprehensive, methodologically rigorous master thesis project that systematically addresses four critical research gaps in personality-adaptive AI for caregiver support. By developing, validating, and deploying an **Adaptive LLM-Based Care Coach** specifically designed for Switzerland's 700,000+ informal caregivers, this thesis makes six interconnected contributions spanning personality psychology, conversational AI, implementation science, and Swiss healthcare contextualization.

The project is feasible within a 25-week master thesis timeframe, leveraging an existing N8N personality detection foundation, open-source technologies, and expert consultation. Rigorous evaluation using standardized instruments, longitudinal follow-up, and fairness auditing establishes reproducible evidence enabling meta-analysis and future research.

Beyond academic contributions, this work provides immediate practical value: a scalable, evidence-based, culturally appropriate support system providing personality-adaptive emotional support and Swiss healthcare knowledge navigation for informal caregivers. The system addresses psychological and informational needs (compassion fatigue management, boundary-setting support, policy navigation) but does not resolve economic challenges caregivers may face. Open science release and implementation toolkit enable Swiss healthcare organizations and international research community to build upon this foundation, extending impact to global caregiver support systems and other vulnerable populations requiring theoretically grounded, safe, and equitable personality-adaptive AI.

Expected outcomes: - Six validated solutions to critical research gaps - Evidence for effectiveness (user satisfaction ≥4.0/5.0, sustained engagement patterns) - Longitudinal durability evidence (3-6 months) - Open-source release enabling replication and extension - Implementation toolkit for Swiss healthcare organizations - Foundation for future RCTs, multilingual expansion, and integration with EHR/Spitex systems

This thesis establishes a new paradigm for **theoretically grounded, personality-aware, safe, and contextually appropriate AI applications in informal caregiving**, with implications extending to mental health support, chronic disease management, and other healthcare domains requiring adaptive, empathetic, and reliable conversational AI systems.